

DMS 613 PROJECT PRESENTATION

Option Valuation Using Fast Fourier
Transforms

INTENDED OUTCOMES

1. Understanding the derivation of the Fourier transform of a modified call price.
2. Exploring why the damping coefficient α is necessary to ensure square integrability of call pricing function.
3. To study how the analytic expression for the Fourier transform is mapped onto the FFT algorithm to achieve real-time pricing.
4. To study VG processes and its role in addressing the biases of standard Black Scholes Model and how is FFT implemented in VG Processes.

INTRODUCTION TO FOURIER METHODS

1. Most of the recent literature in mathematical finance has successfully applied Fourier analysis to determine option prices.
2. Existing methods numerically solve for the Delta and the risk-neutral probability of an option finishing in-the-money.
3. These probability elements are theoretically attractive because they allow an option value to be expressed as a simple combination of the stock price and the strike price as $C = S\Pi_1 - Ke^{-rT}\Pi_2$.
4. These decompositions are often numerically undesirable due to the discontinuity of the payoffs, which can lead to slow convergence and pricing errors

WHY WE NEED FFT?

1. FFT is computationally very efficient and reduces the time complexity to $O(N \log N)$ from $O(N^2)$.
2. The pricing of a European Call Option involves calculating the expected value of payoff under a risk-neutral probability measure.
3. In many advanced models like the Variance Gamma Model have complex PDF but they often have simpler Characteristic Function(ϕ) that is the Fourier transform of the PDF.

WHY TRADITIONAL FFT METHODS FAIL?

1. In traditional methods that solve for Delta and risk-neutral probabilities, the integrand contains a term iu in the denominator. This induces a singularity at the evaluation point $u=0$, making the FFT inapplicable.

$$\Pi_1 = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left(\frac{e^{-iu \ln(K)} \phi_T(u-i)}{iu \phi_T(-i)} \right) du, \quad \Pr(S_T > K) = \Pi_2 = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left(\frac{e^{-iu \ln(K)} \phi_T(u)}{iu} \right) du.$$

2. The initial call value $C_T(k)$ is related to the risk-neutral density $q_T(s)$ by:

$$C_T(k) \equiv \int_k^\infty e^{-rT} (e^s - e^k) q_T(s) ds.$$

3. The initial call value $C_T(k)$ tends toward the initial spot price S_0 as the log-strike k tends to minus infinity.
4. Because the function does not vanish at the negative limit, the call pricing function is not square-integrable, which is a requirement for standard Fourier transforms.

THE DAMPING FACTOR

1. The raw call pricing function $C_T(k)$ tends to the spot price S_0 as the log-strike k tends to minus infinity, meaning it is not square-integrable and lacks a standard Fourier transform.
2. To obtain a square-integrable function, we define the modified call price $c_T(k) \equiv \exp(\alpha k) C_T(k)$ for a positive damping coefficient.
3. By applying the Fourier transform to $c_T(k)$, we derive an expression entirely in terms of the model's characteristic function: $\frac{e^{-rT} \phi_T(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}$.
4. Once the Fourier transform is known, the call price is retrieved using the inverse transform:

$$C_T(k) = \frac{\exp(-\alpha k)}{2\pi} \int_{-\infty}^{\infty} e^{-ivk} \psi_T(v) dv = \frac{\exp(-\alpha k)}{\pi} \int_0^{\infty} e^{-ivk} \psi(v) dv.$$

BOUND ON ALPHA

1. Alpha must be greater than 0 to pull the left tail of the call price down to zero.
2. if Alpha is too high, the right tail starts to explode toward infinity.
3. For the modified transform to be well-behaved, the function $\psi_T(0)$ must be finite.
4. The above condition requires the characteristic function to be finite, which is equivalent to $E[S_T^{\alpha+1}] < \infty$.
5. if we set Alpha=0 (i.e. no damping), the denominator of our transform equation vanishes at $v=0$, creating a singularity that crashes the FFT algorithm.

TRUNCATION ERROR ANALYSIS

1. While the math says we should integrate to infinity, computer can only sum a finite number of points. Truncation is the act of stopping at a high enough value a .
2. The truncation error ϵ is defined as the contribution of the integrand over the omitted interval. Mathematically it can be expressed as:

$$\epsilon(a) = \frac{e^{-\alpha k}}{\pi} \int_a^{\infty} \text{Re}[e^{-ivk} \psi_T(v)] dv$$

3. Taking modulus both the sides and applying appropriate inequalities we get bound on the Fourier transform then the bound back into the integral we get: $a > \frac{\exp(-\alpha k)}{\pi} \frac{\sqrt{A}}{\epsilon}$. We can minimize ϵ by choosing appropriate a .

TIME VALUE PROBLEM

1. As time to maturity T tends to 0 the call value $C_T(k)$ approaches its intrinsic value.
2. This proximity to the payoff function creates a sharp kink in the price function, which translates to high-frequency oscillations in the Fourier domain.
3. For extremely small maturities, the time value of the option begins to approximate the shape of a Dirac delta function centered at $k = \ln(S_0)$.
4. to fix this, we need to stop focusing on the total price and start looking only at the Time Value.

ISOLATION OF TIME VALUE

1. The function $z_T(k)$ is defined as the T-maturity Put price when the strike is out-of-the-money ($k < \ln(S_0)$) and the T-maturity Call price when it is out-of-the-money ($k > \ln(S_0)$).
2. $z_T(k)$ naturally peaks at the at-the-money point ($k = \ln(S_0)$) and declines toward zero in both directions as k tends to plus minus infinity.
3. Assuming $S_0 = 1$, the time value is expressed as:

$$z_T(k) = e^{-rT} \int_{-\infty}^{\infty} [(e^k - e^s)1_{s < k, k < 0} + (e^s - e^k)1_{s > k, k > 0}] q_T(s) ds$$

4. By subtracting the intrinsic value, we are able to remove the kink at the strike price.

HYPERBOLIC DAMPING

1. Although there is no issue regarding the behavior of $z_T(k)$ as k tends to positive or negative infinity.
2. But as $T \rightarrow 0$, $z_T(k)$ behaves like a Dirac delta function at $k = 0$, causing the transform to become wide and oscillatory.
3. To overcome this we use a hyperbolic sine damping function that ensures that the function vanishes at $k=0$.

$$\gamma_T(v) = \int_{-\infty}^{\infty} e^{ivk} \sinh(\alpha k) z_T(k) dk = \frac{\zeta_T(v - i\alpha) - \zeta_T(v + i\alpha)}{2}$$

4. The final time value derived is thus: $z_T(k) = \frac{1}{\sinh(\alpha k)} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ivk} \gamma_T(v) dv$

DISCRETIZATION FOR FFT

1. The FFT computes the following sum for $k = 1, \dots, N$: $w(k) = \sum_{j=1}^N e^{-i \frac{2\pi}{N} (j-1)(k-1)} (j)$
2. To use the FFT algorithm we need to first of all discretize the continuous pricing integral.
3. We can use the Trapezoid Rule by setting $v_j = \eta(j-1)$. Now the call price approximately becomes: $C_T(k) \approx \frac{e^{-\alpha k}}{\pi} \sum_{j=1}^N e^{-i v_j k} \psi_T(v_j) \eta$ and $a = N\eta$.
4. The authors have used a regular spacing of size λ for N values of k : $k_u = -b + \lambda(u-1)$ for $u = 1, \dots, N$.
5. This provides a range from $-b$ to b , where $b = 1/2N\lambda$. Substituting the grids for v_j and k_u into the approximation we get: $C_T(k_u) \approx \frac{e^{-\alpha k_u}}{\pi} \sum_{j=1}^N e^{-i \lambda \eta (j-1)(u-1)} e^{i b v_j} \psi_T(v_j) \eta$
6. To get the form required by the FFT summation, we must impose the following restriction on the grid step sizes: $\lambda \eta = 2\pi / N$.
7. This constraint implies a trade-off: a very small η leads to a large λ , which is not desirable.

SIMPSON'S RULE IMPLEMENTATION

1. To utilize the FFT algorithm, the log-strike spacing (λ) and the integration step (η) must satisfy the Nyquist-like restriction:

$$\lambda\eta = 2\pi/N.$$

2. This restriction implies that a very fine integration grid results in a coarse strike grid, potentially missing detail near at-the-money prices.

3. To overcome this the authors incorporate Simpson's rule weightings into our summation to get:

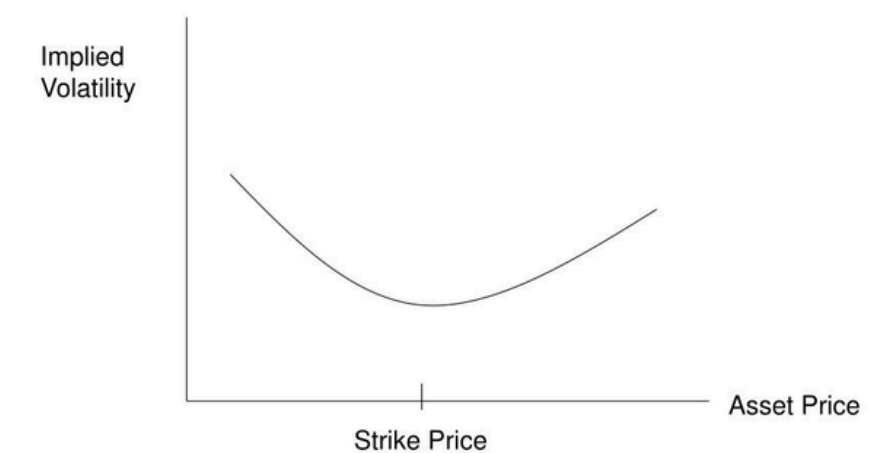
$$C(k_u) = \frac{e^{-\alpha k_u}}{\pi} \sum_{j=1}^N e^{-i\frac{2\pi}{N}(j-1)(u-1)} e^{ibv_j} \psi(v_j) \frac{\eta}{3} [3 + (-1)^j - \delta_{j-1}]$$

4. For out of the money options the process is same we just need to replace $e^{\alpha k}$ by $\sinh(k\alpha)$.

THE VOLATILITY SMILE ISSUE IN BSM

1. The BSM assumes that the volatility of the underlying asset is constant across all strike prices and maturities
2. In real-world markets, plotting implied volatility against strike prices reveals a "U-shaped" curve known as the volatility smile.
3. This curve indicates that BSM systematically misprices out-of-the-money (OTM) options, as the market prices in a higher probability of extreme events than a normal distribution suggests.
4. BSM assumes log-normal returns, which fails to account for the kurtosis and skewness observed in actual financial data.

Volatility Smile



VG PROCESSES

1. The VG process, $X_t(\sigma, \theta, \nu)$, is obtained by evaluating an arithmetic Brownian motion with drift θ and volatility σ at a random time given by a gamma process with a mean rate of 1 and a variance rate of ν .
2. It effectively removes the systematic biases and "smiles" observed in Black-Scholes implied volatilities.
3. While the density function is mathematically complex, the characteristic function for the log-price s_T is simple and analytic, making it ideal for the FFT:
$$\phi_T(u) = \exp[iu(\ln S_0 + (r + \omega)T)] \left(1 - i\theta\nu u + \frac{1}{2}\sigma^2 u^2 \nu\right)^{-T/\nu}$$
4. For the FFT to converge, the damping factor α must satisfy $E[S^{(\alpha+1)}T] < \infty$. For the VG model, this requires:
$$\alpha < \sqrt{\frac{\theta^2}{\sigma^4} + \frac{2}{\sigma^2 \nu}} - \frac{\theta}{\sigma^2} - 1$$

CONCLUSION

1. The FFT framework is a generic solution and not limited to Variance Gamma only it can be used for any model whose characteristic function is analytically known.
2. By using damping factor α and hyperbolic smoothing $\sinh$, we can turn jagged option payoffs into smooth, integrable functions.
3. The "Time Value" approach, utilizing $\sinh$ damping, provided a critical solution for short-dated options, suppressing the oscillations and maintaining precision as $T \rightarrow 0$.

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THANK YOU

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